SereniSuit

A wearable chest-suit platform that monitors physiological signals associated with arousal/stress and provide non-invasive tactile feedback.

AD8232 ECG · ESP32-S3 · DRV8833 haptics · OLED · Wi-Fi + Bluetooth dashboards

PROBLEM STATEMENT

Stress and abnormal heart rates stay invisible until they become emergencies

SereniSuit is a smart wearable chest suit designed to monitor physiological signals such as **heart activity, breathing, skin conductance, and body movement** to identify increased physiological arousal during stressful situations. Based on the detected changes, the system provides **adaptive vibration feedback** to help the user regulate breathing and support a calmer physiological state.

Invisible

Stress raises the heart rate long before a person notices it or asks for help.

Occasional

ECG and BP checks happen in a clinic or a screening camp — not during a normal day.

Passive

Most wearables show a number on the wrist; nothing on the body helps you calm down.

Why this problem matters to us

- People experiencing **anxiety, panic attacks, and severe stress** can undergo sudden changes in heart rate, breathing, sweating, and body movement.
- People with certain **cardiac rhythm disorders (arrhythmias)** may experience abnormal or irregular heartbeats, highlighting the importance of continuous physiological monitoring.
- During stressful or panic situations, people may **not recognize their physiological changes early**, making timely self-regulation difficult.
- This inspired the idea of developing a **wearable suit that continuously monitors physiological signals** such as ECG, respiration, GSR, and movement.
- The project aims to provide **real-time vibration feedback** based on detected physiological changes, supporting the user in calming and regulating their physiological arousal.

IN THE NEWS

The people this problem has already hurt

Hassan, Karnataka · 2025

18 heart-attack deaths in a month

Victims aged about 30–55. The government set up an expert committee under Jayadeva Institute and launched ECG and BP screening camps.

The Week · 1 Jul 2025

Pune · 2024

Anna Sebastian Perayil, 26, chartered accountant

Died four months into her first job at EY; her family blamed the work pressure. The Labour Ministry opened a probe.

Tribune India · Sep 2024

Rajasthan & Maharashtra · 2025

Children collapsing from heart attacks

A 9-year-old girl collapsed at school; a 10-year-old boy collapsed while playing at a Ganpati celebration.

The Week · 22 Oct 2025

Worldwide

Stress at work is a global health issue

WHO and ILO call for new measures: 12 billion workdays lost each year to depression and anxiety.

WHO · 28 Sep 2022

OUR SOLUTION

SereniSuit senses, decides, calms and informs — on the body

SENSE

Chest ECG

Three gel pads and an AD8232 read the heart's electrical signal — the most direct heartbeat measure.

DECIDE

On-chip

ESP32-S3 finds each R-peak (Pan-Tompkins), computes BPM and checks the 60–100 BPM range. No cloud.

CALM

Haptic cue

On DANGER, a DRV8833 drives a vibration motor on the shoulder — a nudge to slow down and breathe.

INFORM

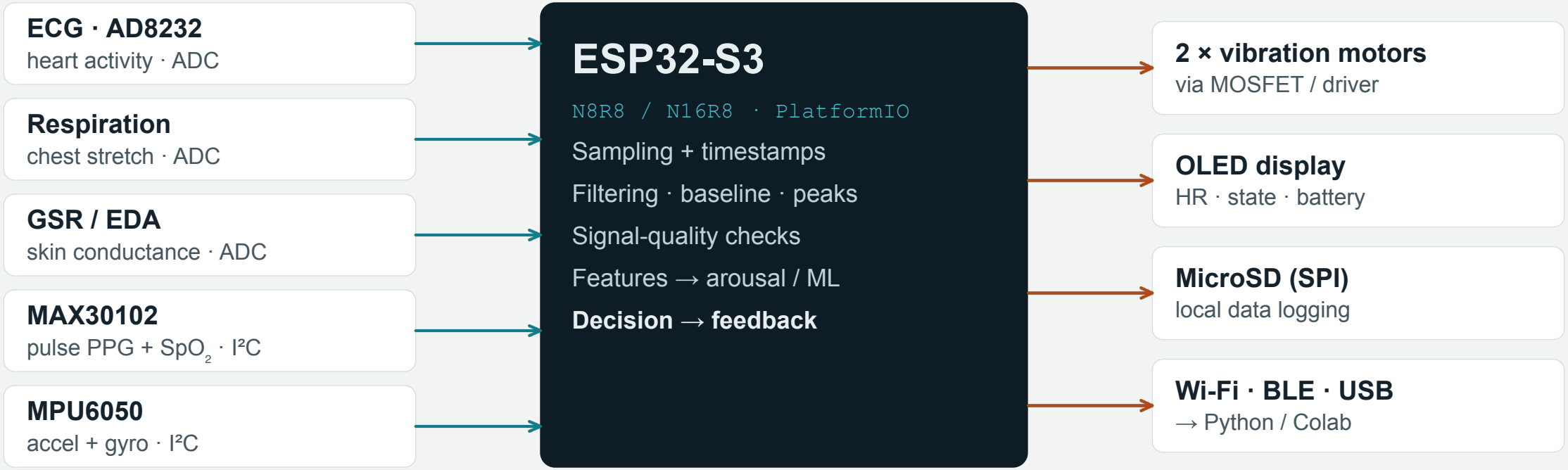
Every screen

OLED on the suit, a laptop website over Wi-Fi and a phone app over Bluetooth, with alerts on each check.

Full vision adds skin conductance (GSR), breathing, PPG and motion so the suit can tell **stress** from **exercise** — Demo 1 proves the core loop with ECG.

SYSTEM ARCHITECTURE

Five sensors in, one brain, four ways out



Power: 1-cell Li-ion / LiPo → TP4056 protected charger → regulated 3.3 V rail → ESP32 + sensors. Never a raw battery on the 3.3 V rail.

RESEARCH · THE PROBLEM AND THE SIGNAL

Papers that tell us what to measure and how

Kim et al. · Psychiatry Investig. 2018

Stress and Heart Rate Variability: a meta-analysis

Main point: Stress measurably shifts the heart's autonomic balance — lower parasympathetic (HF) activity — visible in beat-to-beat intervals.

For SereniSuit: Justifies reading R-R intervals from the chest as a stress signal; HRV features are our next step.

Pan & Tompkins · IEEE TBME 1985

A real-time QRS detection algorithm

Main point: Band-pass → derivative → square → moving-window integral → adaptive threshold; 99.3 % of QRS complexes found on MIT/BIH.

For SereniSuit: Our firmware runs a simplified Pan-Tompkins at 250 Hz to find every R-peak.

Bousseljot et al. · PTB database 1995

PTB Diagnostic ECG Database (PhysioNet)

Main point: A public set of labelled ECG recordings, healthy and myocardial-infarction.

For SereniSuit: Trained our 110 KB beat-shape MLP (0.961 accuracy, 0.985 ROC-AUC on a held-out split).

RESEARCH · THE SOLUTION

Papers that show sensing + feedback can calm people

Azevedo et al. · Scientific Reports 2017

The calming effect of a new wearable device

Main point: 52 people; a heartbeat-like vibration ~20 % slower than their resting heart rate lowered skin-conductance rise and self-reported anxiety before a speech.

For SereniSuit: Evidence that a haptic vibration on the body can calm — the role of our motor.

Goessl et al. · Psychological Medicine 2017

HRV biofeedback for stress and anxiety: a meta-analysis

Main point: 24 studies, 484 people: biofeedback gave a large drop in stress and anxiety (Hedges' g = 0.83 vs control); wearables are promising.

For SereniSuit: Supports the closed loop: measure the heart, feed it back to the user.

Schmidt et al. · ICMI 2018

WESAD: wearable stress and affect detection

Main point: A chest + wrist dataset (ECG, EDA, respiration, motion) from 15 people for stress vs. calm classification.

For SereniSuit: Roadmap for adding GSR, breathing and motion, and a dataset to train a real stress model.

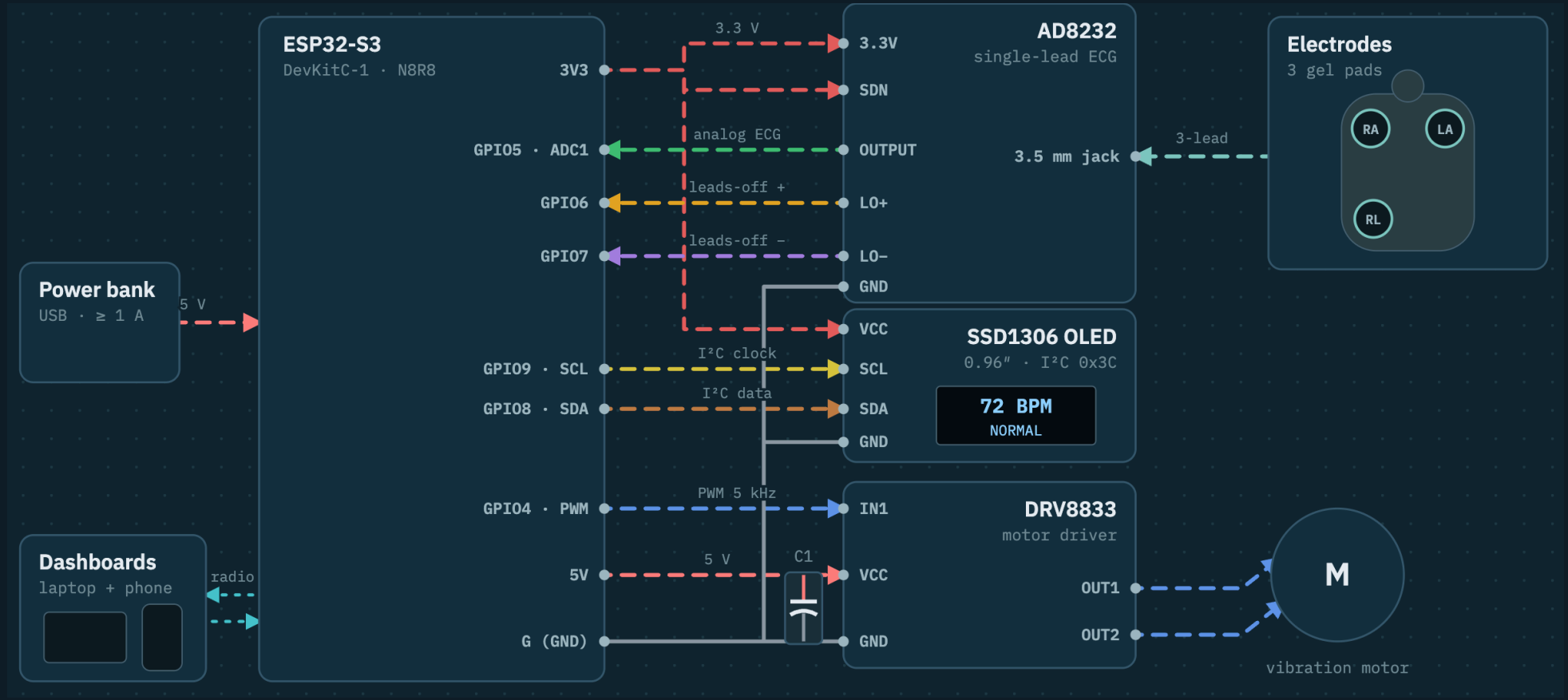
FEATURES

Features of the full SereniSuit

Feature	How it solves the problem	Status
Chest ECG heart rate (AD8232)	Continuous BPM from the heart's own signal, not a clinic visit	Demo 1
On-device decision (ESP32-S3)	NORMAL / DANGER in seconds, no internet or cloud needed	Demo 1
Haptic vibration feedback	A nudge on the body to slow down and breathe	Demo 1
OLED display on the suit	Wearer sees BPM and status instantly, no phone needed	Demo 1
Wi-Fi website + Bluetooth app	Live BPM on laptop and phone for a caretaker or doctor	Demo 1
Alerts on every check	Pop-up, sound and phone buzz — nobody misses a DANGER	Demo 1
Electrode-off detection	Says "attach electrodes" instead of raising a false alarm	Demo 1
Remote on/off + demo BPM	Control and test from any paired screen, hands-off	Demo 1
On-chip TinyML beat model	Screens beat shape on-device, privately, no cloud	Self-test
GSR skin conductance	Sweat response is stress-specific: stress vs exercise	Planned
Respiration sensor	Breathing rate; guides paced breathing to calm down	Planned
MAX30102 PPG / SpO_2	Backup pulse and blood-oxygen from the finger	Planned
MPU6050 motion	Flags movement and activity to avoid false alarms	Planned
Battery + MicroSD logging	All-day wear; logged data personalises the model	Planned

CIRCUIT DIAGRAM

How the signal moves through Demo 1

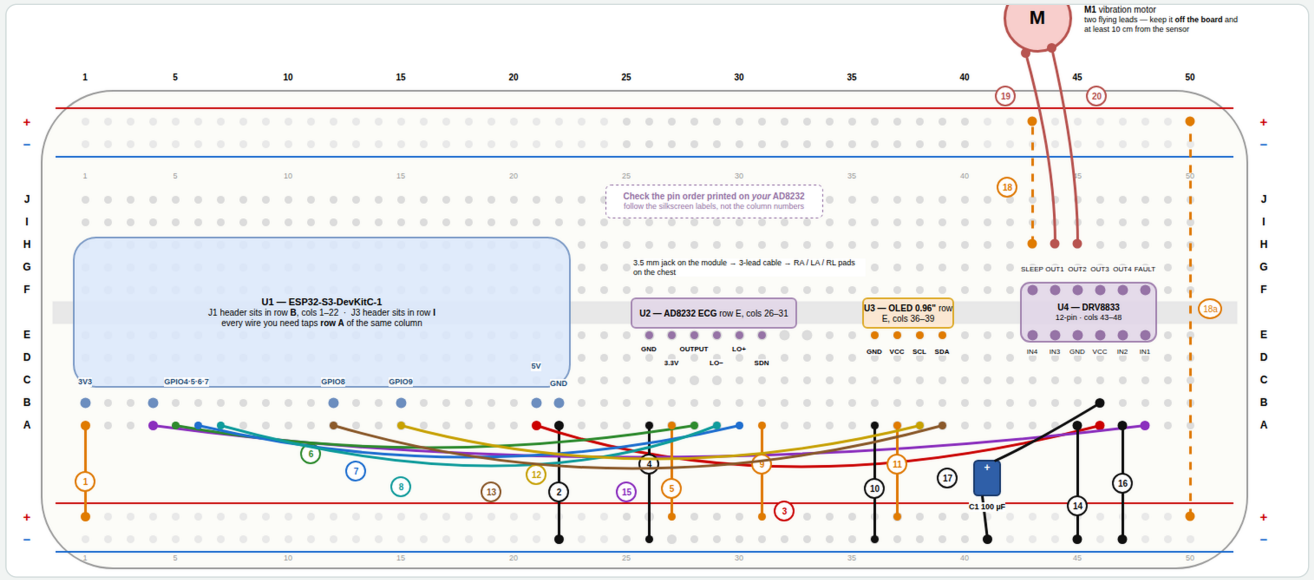


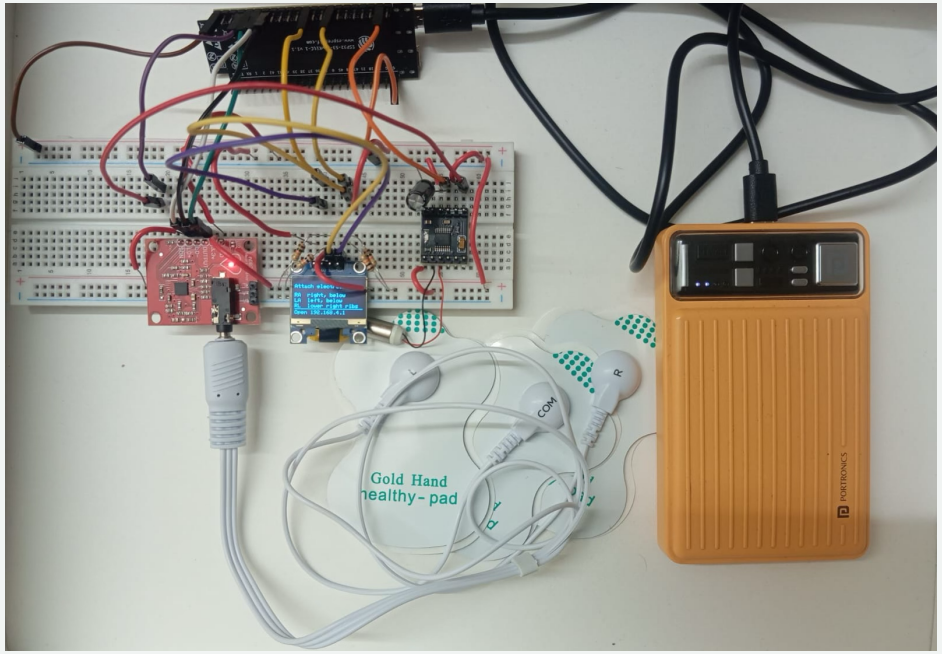
Circuit diagram

Hover version — point at any part or wire: [SereniSuit Circuit Explorer]

BREADBOARD WIRING

MB102 breadboard, hole by hole





Power

ESP32 on cols 1–22

1 3V3 → + rail

2 G → – rail

3 5V → DRV VCC

+ rail = 3.3 V only

AD8232 ECG

row E · cols 26–31

4 GND → –

5 3.3V → +

6 OUT → GPIO5

7 LO+ → GPIO6

8 LO– → GPIO7

9 SDN → +

OLED

row E · cols 36–39

10 GND → –

11 VCC → +

12 SCL → GPIO9

13 SDA → GPIO8

DRV8833 + motor

cols 43–48

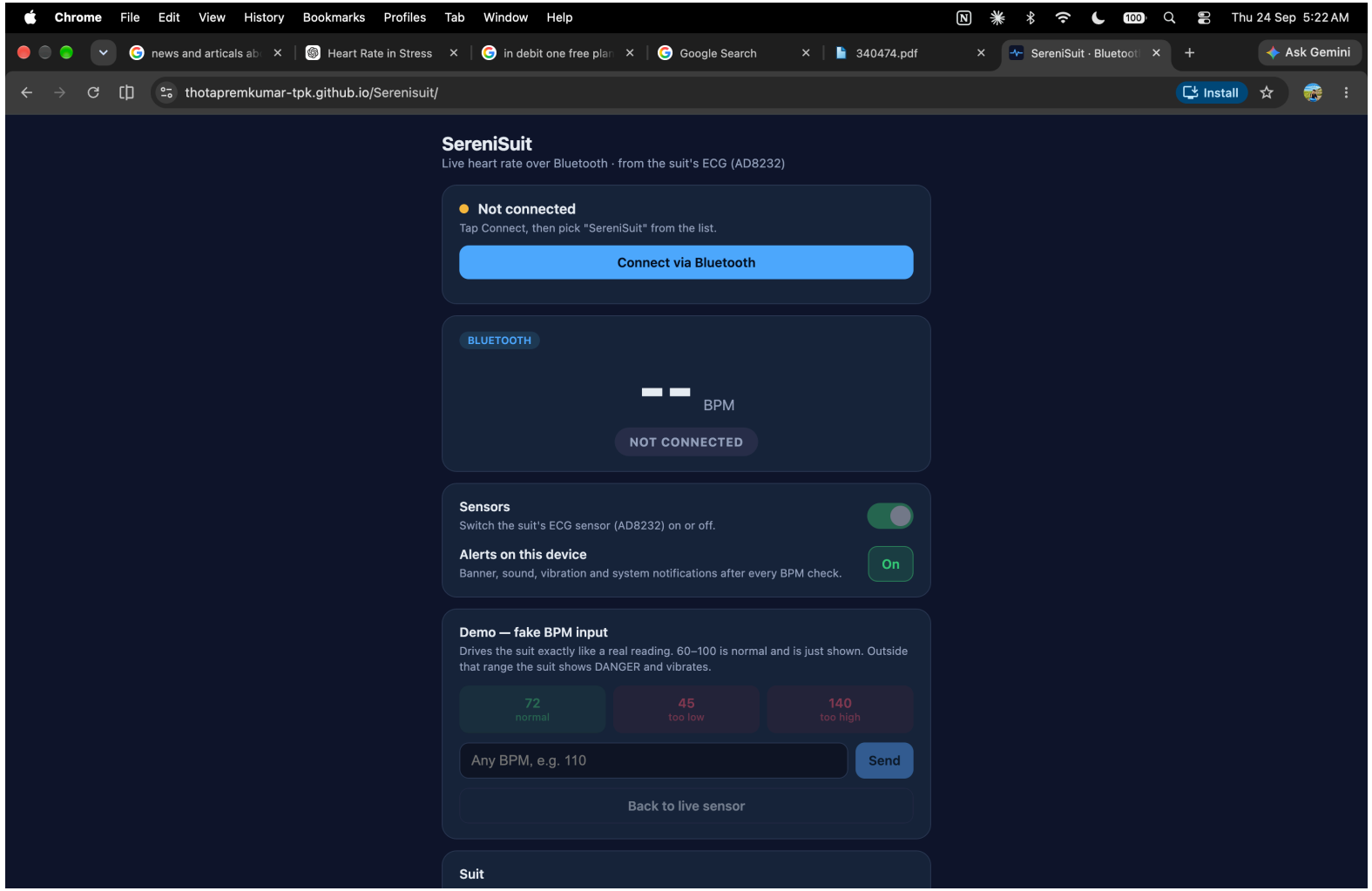
14 GND → –

15 IN1 ← GPIO4

16 IN2 → –

17 C1 100 µF

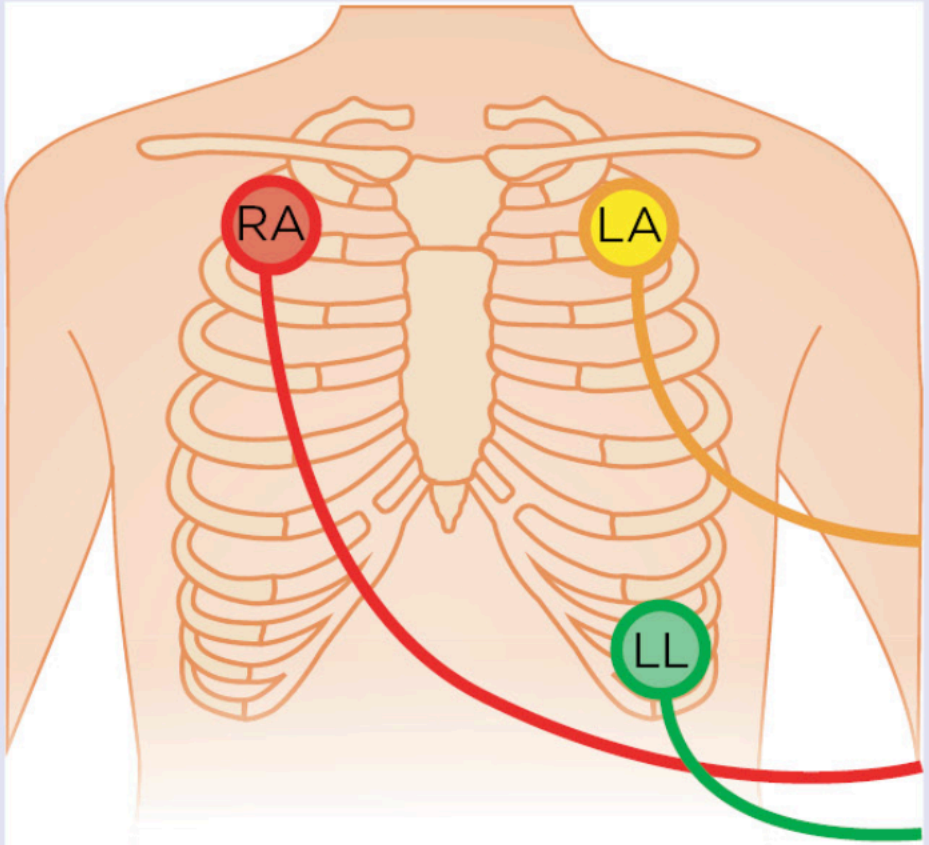
19/20 OUT1/2 → M



Website link : - https://thotapremkumar-tpk.github.io/Serenisuit/

- In this website or app you can connect the device via bluetooth
- You can control the sensors (on or off)
- You will notified when abnormal bpm is recorded
- In this fake BPM input also added for demo purpose

Fig 7. **Three-lead electrocardiogram placement**



Red: electrode placed under right clavicle near right shoulder within the rib cage frame

Yellow: electrode placed under left clavicle near left shoulder within the rib cage frame

Green: electrode placed on the left side below pectoral muscles lower edge of left rib cage

Thank you

SereniSuit reads heart rate from a single-lead ECG and responds on the body. It is a research prototype — not blood pressure, and not a diagnostic device.

Questions?